HARDWARE DOCUMENT

Project: ECSE 211 Search & Retrieval Robot

Task: Overall hardware and structural design

Document Version Number: 2.6

Date: March 10, 2020

Author: Yazdan Zinati

Edit History:

Date	Version	Author(s)	Changes
Feb 25	1.0	Yazdan Zinati	First draft
Feb 25	1.1	Yazdan Zinati	Elaborated sections 1.0 - 8.0
Feb 25	1.2	Yazdan Zinati	Elaborated section 9.0
Feb 25	1.2.1	Andy Li William Bouchard	Improved formatting and table of contents
Mar 8	2.0	Yazdan Zinati	Created sections 10.0 and 11.0
Mar 9	2.1	Yazdan Zinati	Elaborated sections 9.1, 9.1.2, 9.3
Mar 10	2.2	Yazdan Zinati	Created sections 9.4 - 9.7
Mar 10	2.3	Trevor Tabah	Elaborated section 9.2
Mar 10	2.4	Andy Li	Updated formatting and editing
Mar 10	2.5	Trevor Tabah	Elaborated section 9.2 and formatting
Mar 10	2.6	Yazdan Zinati	Elaborated Section 1.0

1.0 – TABLE OF CONTENTS

2.0 – HARDWARE DESIGN PROCESS **3**

3.0 – PURPOSE **3**

4.0 – RESOURCES **3**

5.0 – DESIGN CONSTRAINTS **3**

6.0 – RESEARCH AND DEVELOPMENT **3**

7.0 – DESIRED FEATURES AND REQUIRED COMPONENTS **4**

8.0 – PRELIMINARY DESIGN **5**

8.1 – COMPONENT COUNT AND PLACEMENT 5

8.2 – PRELIMINARY DESIGNS 6

8.2.1 – PRELIMINARY DESIGN 1 6

8.2.2 – PRELIMINARY DESIGN 2 8

8.2.3 – PRELIMINARY DESIGN 3 10

9.0 – FIRST DRAFT OF CHOSEN DESIGN **11**

9.1 – DESIGN DECISIONS 12

9.2 – DESCRIPTION 12

9.2.1 - MAIN COMPONENTS

9.3 – FEATURES 13

9.4 – PICTURES 14

9.5 – EVALUATION TABLE 17

9.6 – FUTURE IMPROVEMENTS 17

9.7 – TIMELINE OF DESIGN MODIFICATION 19

10.0 – TESTS **19**

11.0 – GLOSSARY OF TERMS **19**

2.0 – HARDWARE DESIGN PROCESS

We started the design process of the robot based on three important documents: the System Document, which defines the available hardware, its capabilities and limitations, and key features (e.g., reusability and compatibility); the Requirements Document, which defines the purpose, constraints, and desired features of the robot; and the Constraints Document, which shows the limitations we must consider in the hardware design.

We created the preliminary designs based on the design features, taking into account the constraints and limitations we must work with. We then expect to evaluate and compare the relative advantages of each design to narrow down our ideas to the preferred design of the hardware. However, even this preferred design may be later modified to resolve unexpected problems in the future.

3.0 – PURPOSE

For a detailed discussion of the purpose, refer to Section 2.0 of the Requirements Document.

4.0 – RESOURCES

The resources are those found in the three Mindstorms EV3 kits, the contents of which are listed in Section 3.0 of the System Document.

5.0 – DESIGN CONSTRAINTS

The design constraints are listed in the Constraints Document.

6.0 – RESEARCH AND DEVELOPMENT

Since the beginning of this course, we have completed five labs that address different elements of our final design; thus, the hardware design from the previous labs is relevant to our final project. Since the first four labs were completed in smaller labs than Lab 5, we were able to integrate components from each lab in the hardware design for the final lab (Figure 1). We expect to use elements from this hardware design for the project, as we plan to use a similar array of sensors and motors on the final design.

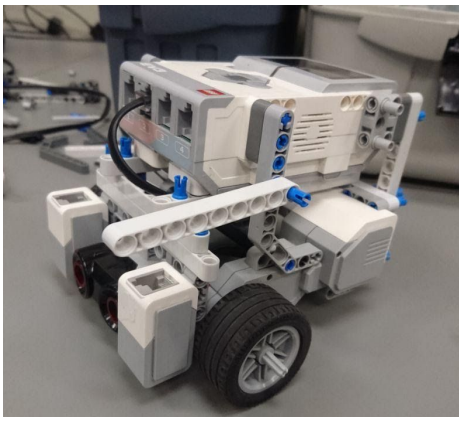


Figure 1: Lab 5 base robot design

7.0 – DESIRED FEATURES AND REQUIRED COMPONENTS

Localization:

- 1 ultrasonic sensor
- 1-2 light sensors (only if light localization is used)

Odometer error correction:

- 2 light sensors

Navigation:

- 2 large motors
- 1 steel ball bearing

Obstacle avoidance:

- 1-2 ultrasonic sensors

Hooking mechanism:

- 1-2 medium motors

8.0 – PRELIMINARY DESIGN

8.1 – COMPONENT COUNT AND PLACEMENT

Ultrasonic sensors

Placement	Advantages	Disadvantages
Bottom, center (1 sensor)	Detects low obstacles Localizer code can be reused	Increases width, wheelbase
Top, center (1 sensor)	Reduces width, wheelbase Localizer code can be reused	Increases robot height Only detects tall obstacles Reduced obstacle detection
Top, forward (2 sensors)	Reduces width, wheelbase Improved obstacle detection	Increases robot height Only detects tall obstacles Localizer code must be rewritten
Facing outward (2 sensors)	Improved obstacle detection	Localizer code must be rewritten Increases robot width

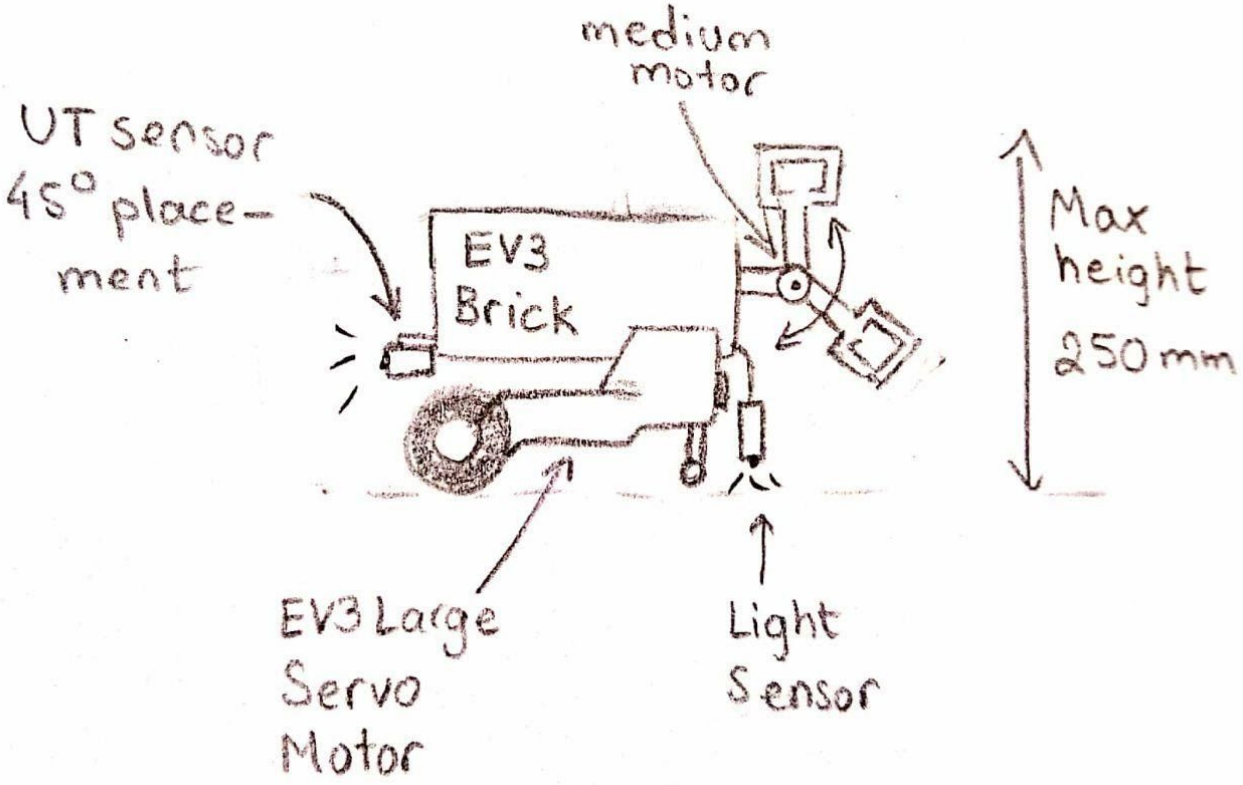
Light sensors

Placement	Advantages	Disadvantages
None	Reduces robot width No coding necessary	No odometer correction
Back of robot (1 sensor)	Reduces robot width Localizer code can be reused	Robot must spin to correct odometer
Sides of robot (2 sensors)	Quick odometer correction	Increases robot width, power consumption Localizer code must be rewritten

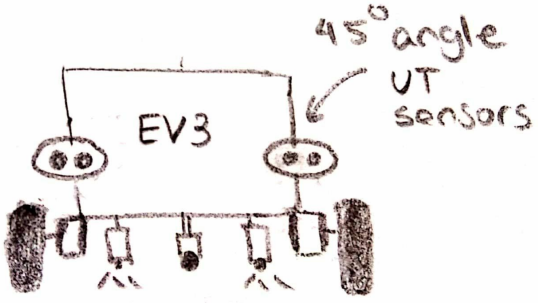
8.2 – PRELIMINARY DESIGNS

8.2.1 – PRELIMINARY DESIGN 1

Side View



Front View



This design consists of two ultrasonic sensors, each placed at a 45 degree angle in the front of the robot, for object detection and ultrasonic localization. Two light sensors, mounted in the back, can be used for odometer correction and light localization. Placing the light sensors in the back reduces the width of the robot. Other variations of this robot could have bottom mounted motors to further reduce the width.

Components:

- 2 wheels
- 2 large EV3 motors
- 1 medium motor
- 2 ultrasonic sensors
- 2 light/colour sensors

Advantages:

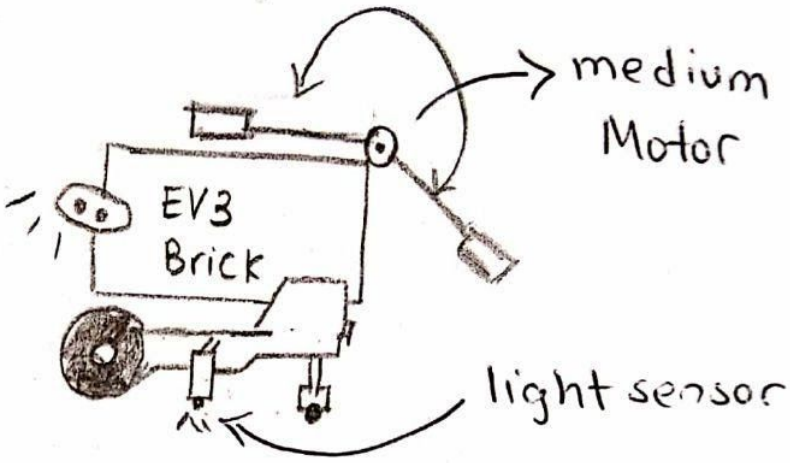
- Relative simplicity
- Two light sensors improve odometer correction
- Reduced robot width

Disadvantages:

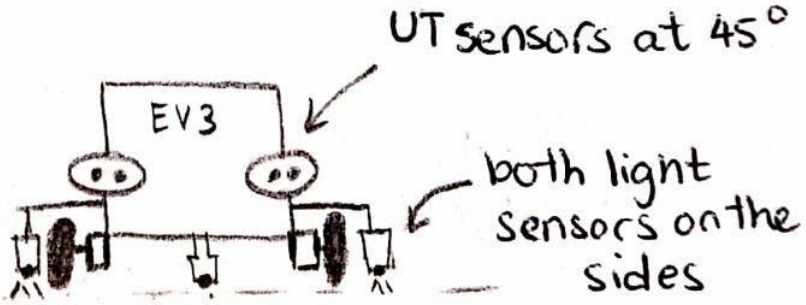
- Motor at back of robot adds weight
- Ultrasonic localization requires new code
- Small distance between light sensors reduces accuracy

8.2.2 – PRELIMINARY DESIGN 2

Side View



Front View



This design contains two ultrasonic sensors, each placed at a 45 degree angle in the front of the robot; the two sensors are used for both object detection and ultrasonic localization. Two light sensors, which can be used for odometer correction and light localization, are also mounted on the sides. The large distance between the side mounting light sensors makes detecting the odometer error easier. However, the robot is also wider, which could cause issues in the tunnel or when turning.

Components

- 2 wheels
- 2 large EV3 motors
- 1 medium motor
- 2 ultrasonic sensors
- 2 light/colour sensors

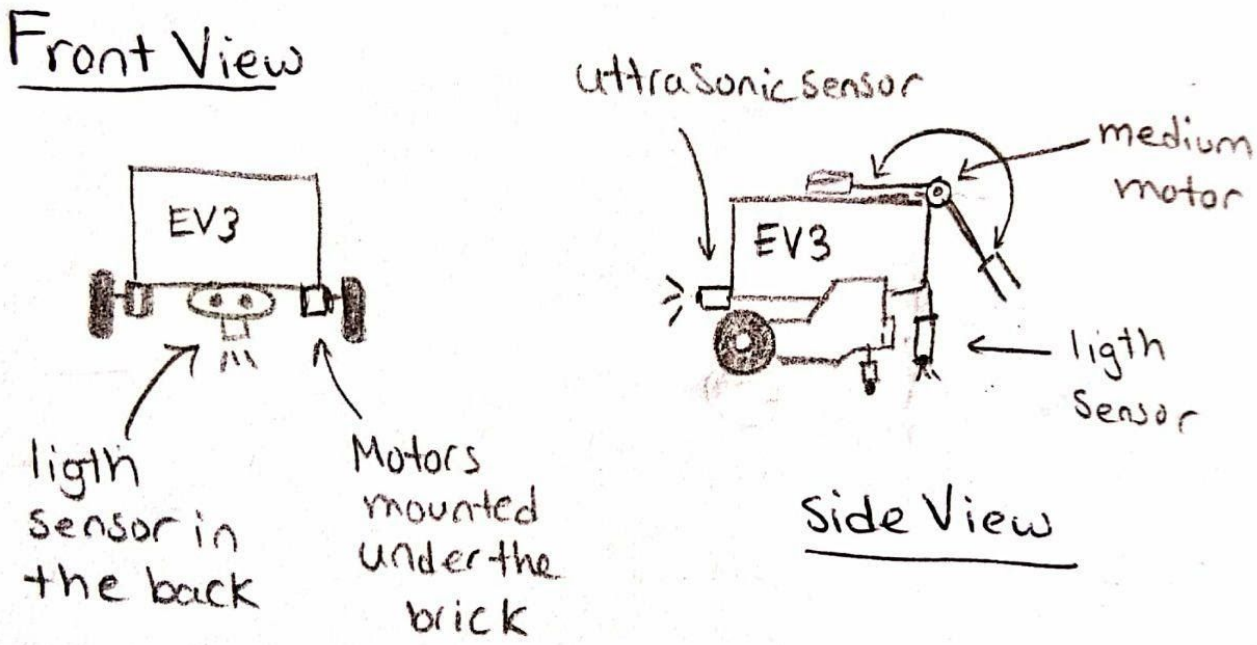
Advantages:

- Two light sensors improve odometer correction, reduce drift
- Distance between sensors increases sensor accuracy
- Improved front-back weight balance

Disadvantages:

- Width of robot reduces maneuverability
- Ultrasonic localization requires new code

8.2.3 – PRELIMINARY DESIGN 3



This design resembles the robot we used in Lab 5. It only has one (front-mounted) ultrasonic sensor and one (rear-mounted) light sensor. The motors and ultrasonic sensor are mounted on the bottom of the brick . This allows the robot to be only slightly wider than the EV3 brick itself.

Components:

- 2 wheels
- 2 large EV3 motors
- 1 medium motor
- 1 ultrasonic sensor
- 1 light/color sensor

Advantages:

- Relative simplicity
- Reduced robt width
- Localization code can be reused
- Improved front-back weight balance

Disadvantages:

- Ultrasonic sensor has reduced field of view
- Light localization requires robot to spin

9.0 – FIRST DRAFT OF CHOSEN DESIGN

9.1 – DESIGN DECISIONS

For the final design of the robot, we opted for a combination of preliminary design 2 and 3. At first we chose to use preliminary design 2 as a base for our final design as it was the most promising out of the 3 designs. However after a meeting with the software lead we realised the possibility of obstacle detection using a light sensor (as all the obstacles are white). Therefore our design goals (refer to section 9.3 - Features) could be met with a simpler design using only one ultrasonic sensor centered in the front. This decision was beneficial in many ways as it makes the software implementation of the ultrasonic localisation from previous labs, reusable and further simplifies our design. The robot width and height are respectively 155 mm and 149 mm which easily fits in the tunnel (Width: 213 mm, Height: 250 mm , refer to Figure 3,4 and 5). This leaves us with some space for possible errors in navigation.

9.2 – DESCRIPTION

Our robot has an ultrasonic sensor facing forward to allow for the sensing of the rings on the course and ultrasonic localization. The light sensor above the ultrasonic sensor will allow for the rings' colours to be detected. It is also the right height to detect the colour of the trailer. On either side of the ultrasonic sensor and between the wheels lie two more light sensors. These will be used for light localization when travelling throughout the course. The purpose of having two light sensors is to allow for the proper adjustment when crossing lines. For example, in some cases when the robot must travel perpendicular to the lines of the course, the wheels may cross the lines at different times which would imply that the robot is off-course. The light sensor would recognise this and adjust the robot's trajectory. The rear of the robot has a "claw" and a medium motor. These two apparatuses will be used together when the trailer is detected. Once the robot sees the trailer, it will adjust itself to the proper distance and rotate 180 degrees. This rotation will allow for the motor to close the claws and latch on to the trailer. The motor will make sure the claw remains closed until the goal is complete. The robot can then travel with the trailer behind it. The two wheels are on the sides of the two light sensors. There is a low-friction metal ball on the rear to allow for better movement and balance. In total, four sensor slots and three motor slots will be occupied.

9.2.1 - MAIN COMPONENTS

- 2 wheels
- 2 large EV3 motors
- 1 medium motor
- 1 ultrasonic sensors
- 3 light/colour sensors

9.3 – FEATURES

- Localization :
    - Both Ultrasonic and Light localization possible
    - A combination of Ultrasonic and Light localization could be improved for higher localization accuracy
- Navigation :
    - Two light sensors in the front allow odometer error correction during navigation by giving feedback, reducing navigation error over long distances
    - The overall small size of the robot allows sharp turn and reduces the turning radius
    - Obstacle avoidance using color detection with the color/light sensor mounted on the top
- Search and Rescue :
    - Both the Ultrasonic sensor and the Light sensors mounted on the top could be used to search for the stranded vehicle
    - Retractable claw design keeps the size of the robot to a minimum (refer to Figure 6)

9.4 – PICTURES

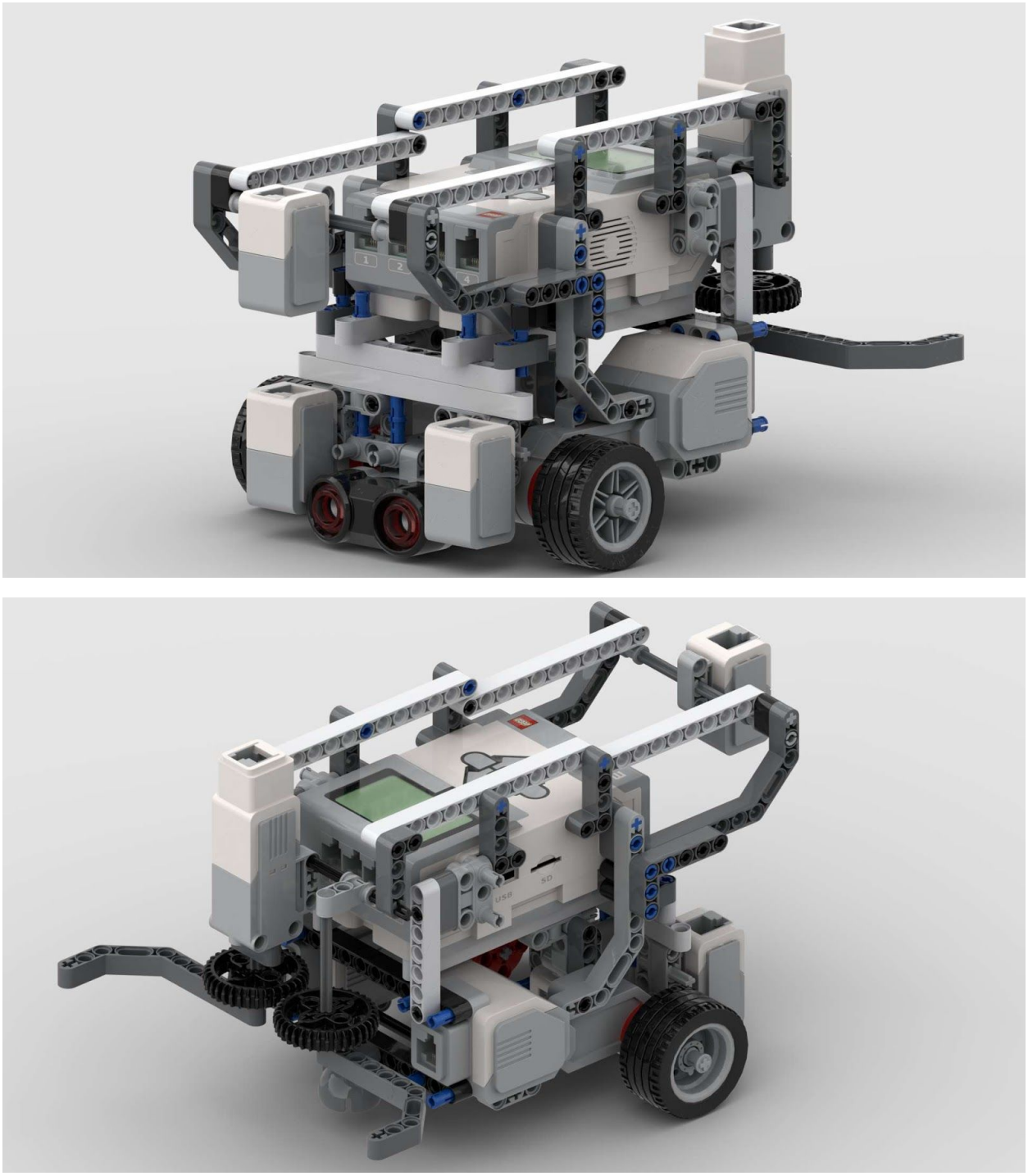


Figure 2 - Overall Design of the robot

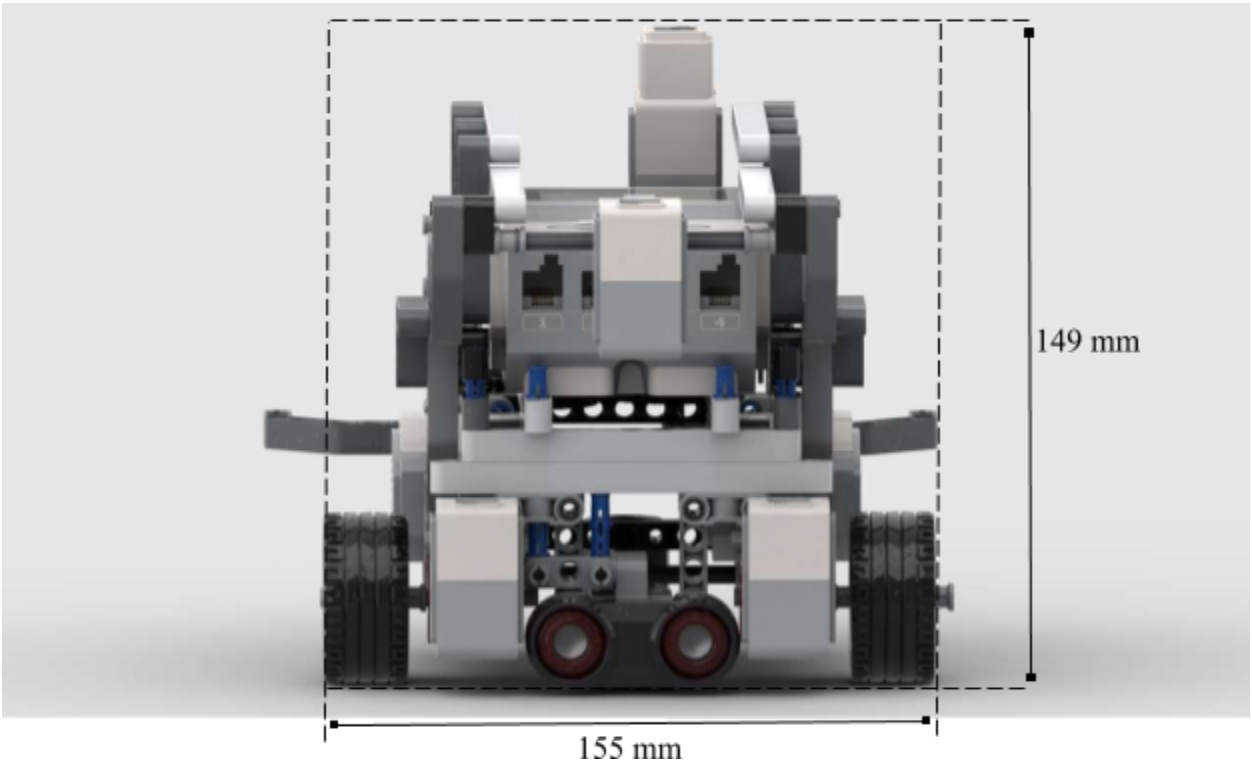


Figure 3 - Front View

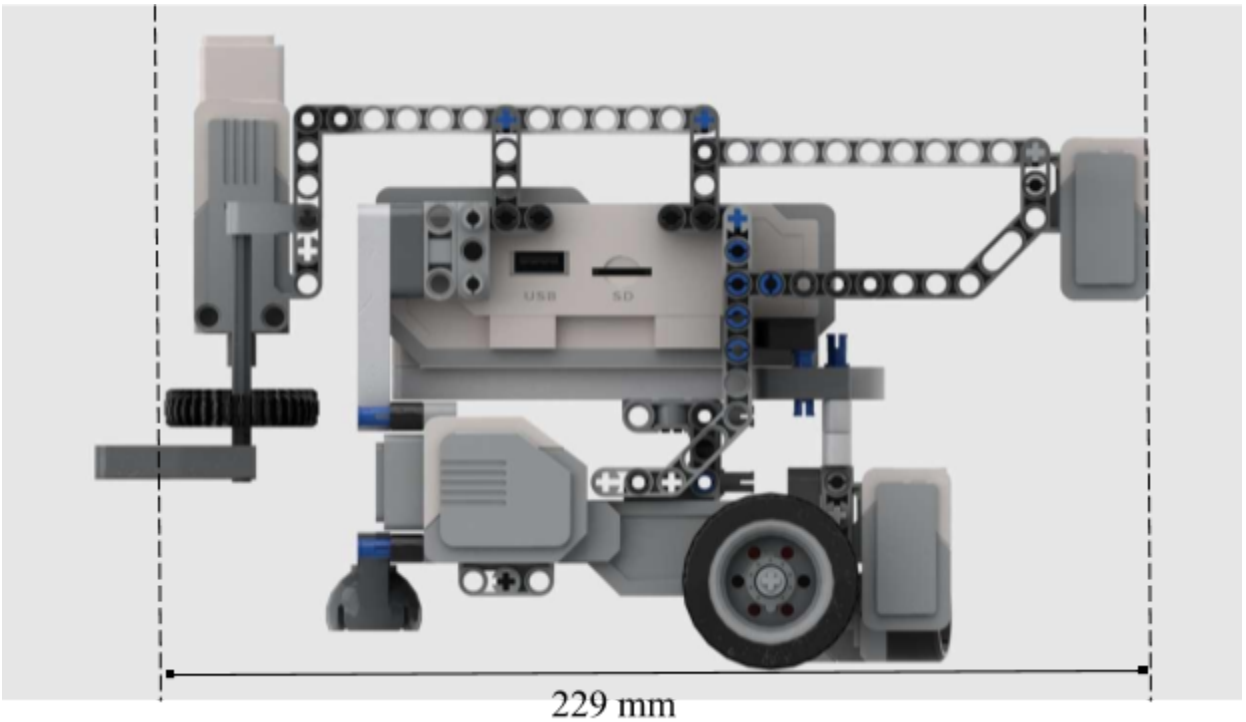


Figure 4 - Side View

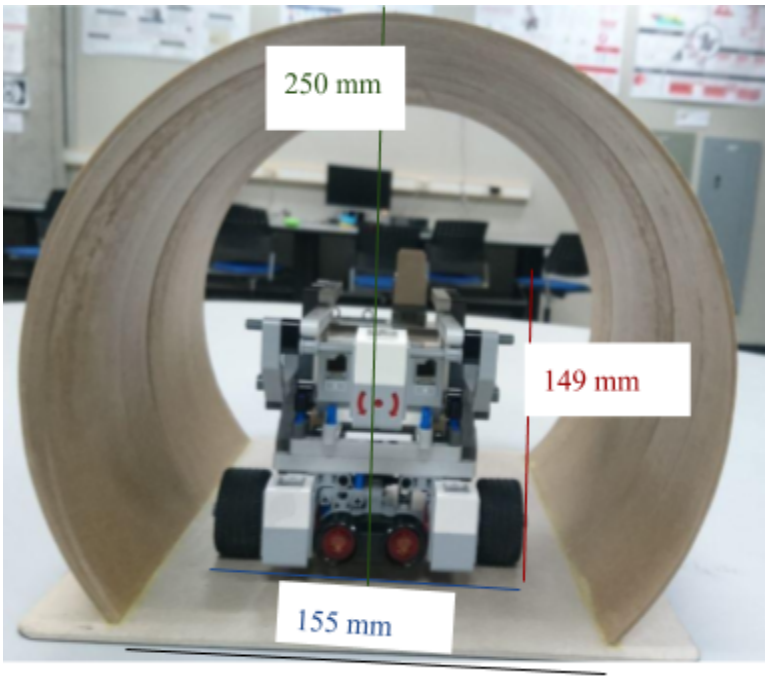


Figure 5 - Robot vs Tunnel dimensions

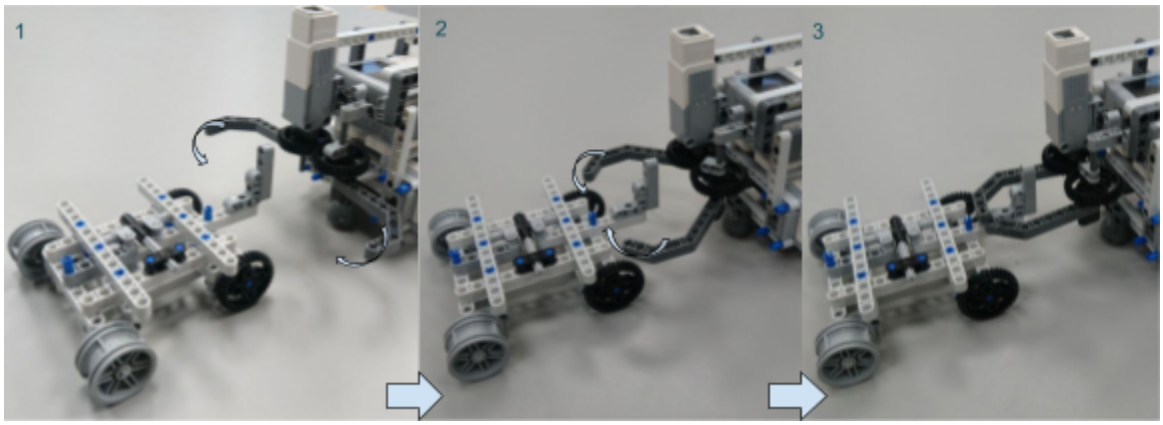


Figure 6 - Pickup procedure of the robot

9.5 – EVALUATION TABLE

Advantages	Disadvantages
Dimensions of the robot within constraints (tunnel size)	Battery not easily accessible
The robot is narrow	The claw and grabbing mechanism may be less consistent
The robot is mostly stable	The pickup claw is not very stable
Two colour sensors for drift correction	A medium motor is heavy - adds weight on the rear, reducing stability

9.6 – FUTURE IMPROVEMENTS

The most notable weakness of our current design is the pickup claw. This part of the robot is less stable than the other components. This reduces the consistency of the robot, as can be seen in Bricklink Studio's stability test (Figure 7). This weakness appears to be due to the axle between the motor and the gears, which does not move freely enough to transmit motion effectively. Another issue to be addressed is the added weight caused by the medium motor responsible for the pickup mechanism. The effects of this weight imbalance can be reduced by moving the motor closer to the center of the robot or adding counterweighting to its front. Finally, it would be useful to swap the battery to charge or change for another one would be useful. However, this would require modifications in the base of the robot and would likely increase its width, reducing maneuverability on the course.

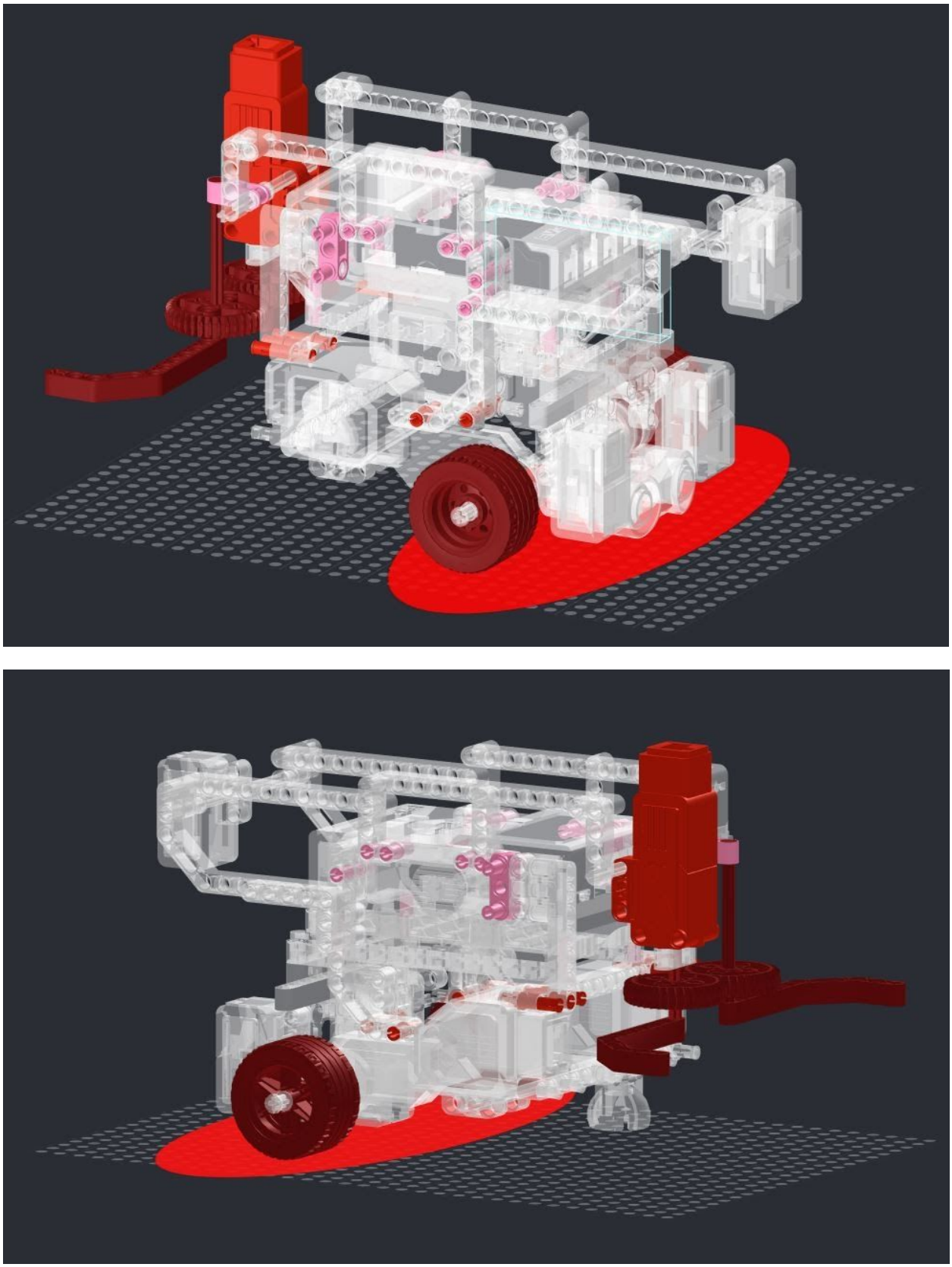


Figure 7 - Stability of the current design (darker shades of red show less stable components)

9.7 – TIMELINE OF DESIGN MODIFICATION

9.7.1 – ROBOT V1.1

(Any future hardware modifications will be recorded in this section)

10.0 – TESTS

Refer to the Testing Document for a detailed description of the testing procedure and its results.

11.0 – GLOSSARY OF TERMS

1. *EV3:* The latest version of the Lego robotics kit (including the processor brick), the base on which we are building our robot.
2. *LDD File*: A Lego Digital Designer file, the software used for modeling of Lego creations on the computer
3. *LXF/LDR*: standard file extension for Lego CAD programs that allow the user to create virtual models and scenes. It can also be used to document models and physical builds.
4. *Bricklink Studio:* Bricklink Studio is an unofficial Lego designer program with additional features not available in Lego Digital Designer, including high-quality 3D rendering, model animation, and model stability analysis.